

Double-Bundle Medial Patellofemoral Ligament Reconstruction Using Free Rectus Femoris Autograft

Samo Novak, M.D.,  and Matic Povh, M.D.* 

Abstract: The rectus femoris tendon autograft represents a relatively new graft option, primarily used in anterior cruciate ligament revision surgery. In this technical note, we describe a technique for medial patellofemoral ligament reconstruction using a double-bundle rectus femoris autograft. This approach combines the favorable biomechanical characteristics of the rectus femoris tendon—providing greater tensile strength and load-to-failure capacity than hamstring autografts—with the anatomical precision of a free graft reconstruction. Unlike the classical quadriceps tendon medial patellofemoral ligament technique, which preserves the native quadriceps insertion at the superior pole of the patella and thus deviates from true medial patellofemoral ligament anatomy, our method allows an anatomic patellar fixation.

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Recurrent patellar instability is a common problem, particularly in adolescents and young adults. Most first-time patellar dislocations occur between the ages of 10 and 19 years, with more than 50% associated with sports-related trauma.¹⁻⁴

Surgical treatment is therefore often indicated, with soft-tissue procedures and realignment techniques favored over osseous procedures.⁵ Medial patellofemoral ligament (MPFL) reconstruction has become the most widely accepted surgical option for addressing recurrent patellar instability. Autografts are preferred, most commonly using semitendinosus or gracilis tendons, though some authors advocate for the quadriceps tendon.⁶⁻⁸ However, harvesting the distal part of the quadriceps tendon could possibly lead to altering patellar kinematics and may increase the risk of patellofemoral arthrosis.

The rectus femoris tendon has recently been explored as a potential graft source in anterior cruciate ligament reconstruction, showing favorable biomechanical properties.^{9,10} In this article, we present a technique for MPFL

reconstruction using the free rectus femoris tendon. This approach preserves the graft's native anatomical alignment and provides a more robust stress profile than hamstring autografts while avoiding the kinematic disadvantages associated with quadriceps tendon harvest.

SURGICAL TECHNIQUE

Patient Positioning

The patient is placed in the supine position, and a pneumatic tourniquet is inflated. Standard diagnostic arthroscopy reveals grade II osteochondral damage on the patella. This lesion is treated with radiofrequency cold ablation (Smith & Nephew, Austin, TX). No additional intra-articular pathology is identified. After completion of the diagnostic arthroscopy, attention is turned to autograft harvesting.

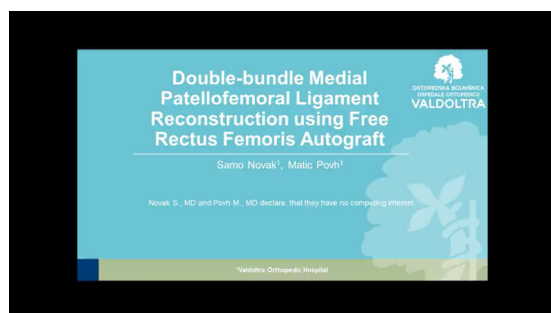
From the Valdoltra Orthopaedic Hospital, Ankaran, Slovenia.

*Address correspondence to Matic Povh, M.D., Orthopaedic Hospital Valdoltra, Jadranska cesta 31, 6280 Ankaran, Slovenia.
E-mail: matic.povh@ob-valdoltra.si

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VIDEO 1 Medial patellofemoral ligament reconstruction using a free rectus femoris autograft in the right knee (supine position). The video shows graft harvest from the rectus femoris tendon, graft preparation with whipstitching of both ends, placement of suture anchors in the patella, subsequent identification of Schöttle's point under fluoroscopic guidance, and final graft fixation with appropriate tensioning to restore patellar stability. When shown, magnetic resonance imaging images of the right knee are presented in the sagittal plane from the medial side to illustrate relevant anatomical landmarks. (AT, adductor tubercle; ME, medial epicondyle; P, patella.) Video content can be viewed at <https://doi.org/10.1002/atn2.70263>.

Graft Harvesting

A 4-cm longitudinal incision is made approximately 3 cm proximal to the superior border of the patella and slightly lateral (Figure 1). This approach minimizes the risk of injury to the vastus medialis muscle and improves visualization.

After subcutaneous tissue dissection, two parallel incisions are made approximately 4 cm proximal to the superior patellar border, spaced 10 mm apart and extending to a depth of 2 mm. A blunt curved dissector is passed from 1 incision to the other, elevating the rectus femoris tendon.

The graft is transected and secured with a size 2 holding suture (Figure 2). Dissection is continued proximally until approximately 8 cm of elastic graft is obtained. Maintaining graft elasticity is crucial; a nonelastic graft may result in distal transection and inadequate length.

After dissecting the initial 8 cm, a closed stripper with an inner diameter of 7 mm (Richard Wolf GmbH, Knittlingen, Germany) is used to complete the graft harvest. The leg is positioned in near extension to facilitate harvesting (Video 1).

The graft is cleared and prepared for double-bundle technique, with both ends whipstitched over using a No. 2 nonabsorbable PowerKnot suture (Medacta International SA, Castel San Pietro, Switzerland) (Figure 3, Video 1).

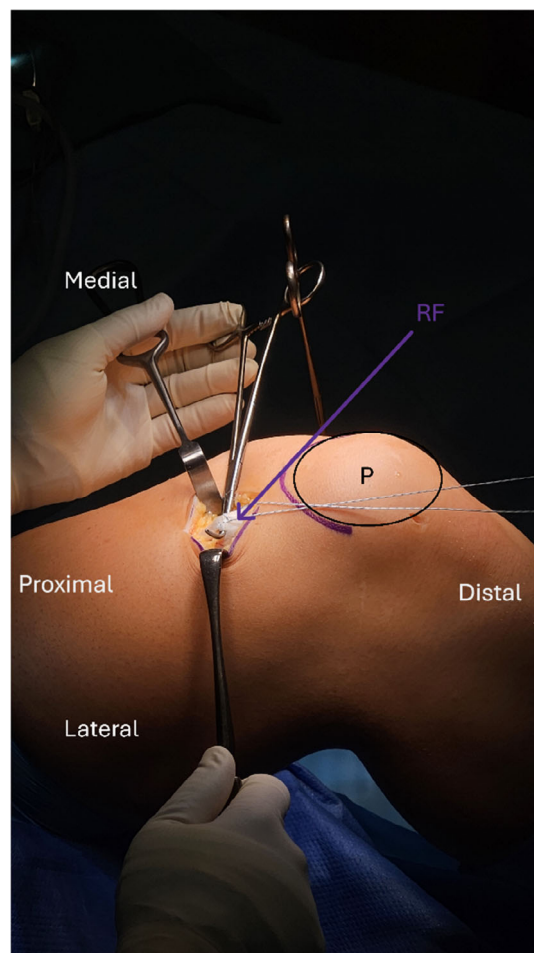


FIGURE 1 Intraoperative photograph of the right knee (supine position), showing a vertical skin incision located approximately 3 cm from the superior border of the P and slightly lateral. A blunt curved dissector is used to separate the RF tendon from the remaining quadriceps tendon. (P, patella; RF, rectus femoris.)

Femoral and Patellar Fixation

The superior border of the patella is identified, and a 2-cm incision is made along its proximal medial third. The bony surface is decorticated, and 3 all-suture anchors Q-Fix 1.8 Mini (Smith & Nephew, Andover, MA) are inserted (Figure 4, Video 1).

Then, a 2-cm skin incision is made between the medial epicondyle and adductor tubercle (Figure 5). Under fluoroscopic guidance, Schöttle's point is identified between these landmarks (Figure 6). A K-wire is drilled through this point.

A blunt dissector is used to create a passage to the femoral fixation site between the second and third layer from the patella. The graft is then drawn through this plane and secured to the patella with sliding knots. Femoral fixation is performed by using a suture-pulling system. With the knee flexed between 45° and 60°, the

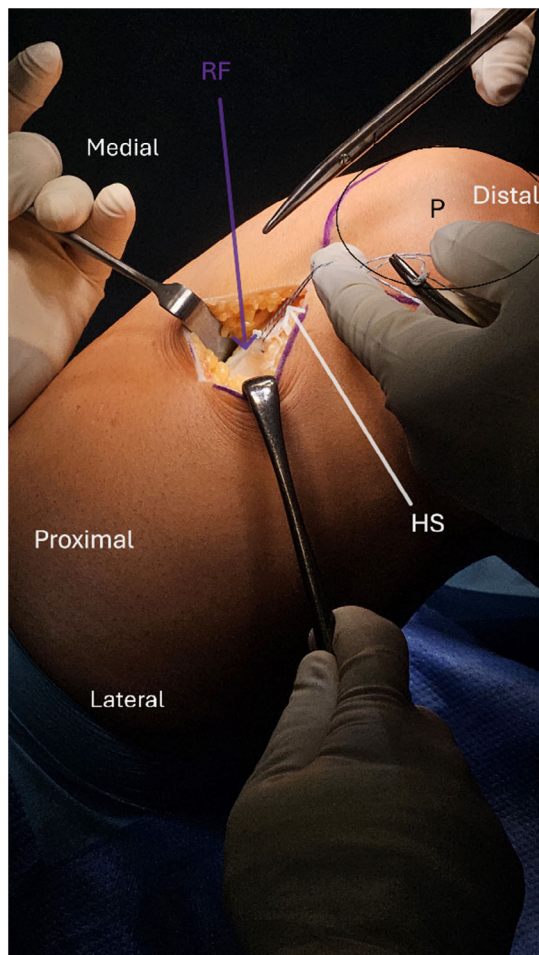


FIGURE 2 Intraoperative photograph of the right knee (supine position), showing graft harvest at the distal end of the skin incision. The graft is detached and secured with a robust HS. (HS, holding suture; P, patella; RF, rectus femoris.)

graft is tensioned until the lateral patellar facet and trochlea are perfectly aligned. Fixation is achieved with a 7×23 mm bioabsorbable interference screw (Milagro, DePuy Mitek, Raynham, MA). Patellar tracking is assessed through the full range of motion (Video 1).

Finally, a second-look arthroscopy confirms that the reconstructed MPFL is extra-articular. Pearls and pitfalls of MPFL reconstruction using a rectus femoris autograft are summarized in Table 1.

Postoperative Protocol

The patient is fitted with an extension knee brace. Flexion is restricted to 90° during the first 6 weeks. Immediate full weight-bearing is permitted, with crutches used as needed for ambulation support.

Return to sports is considered at approximately 6 months, starting with low-intensity activities and progressing to full participation in all sports by 9 months.



FIGURE 3 Intraoperative photograph of the harvested autograft with both ends whipstitched using a No. 2 nonabsorbable PowerKnot suture (Medacta International SA, Castel San Pietro, Switzerland) preparing the graft for fixation. (RF, rectus femoris.)

DISCUSSION

This technical note outlines a surgical technique for double-bundle MPFL reconstruction using a free rectus femoris tendon autograft. Although this graft has been primarily described in the context of anterior cruciate ligament revision procedures, its inherent advantages—particularly sufficient length and straightforward harvesting once key technical considerations are mastered—make it an appealing option for MPFL reconstruction.⁹

We consider the rectus femoris autograft a valuable alternative because it preserves the hamstrings for possible future anterior cruciate ligament reconstruction if required. Unlike hamstring autograft harvesting, it does not compromise medial knee stability.¹¹ Compared with standard partial-thickness graft and pedicled graft techniques, this approach reduces the likelihood of intra-articular entry during harvest. With this technique, the main mixed layers at the proximal patellar insertion of quadriceps tendon are preserved, potentially mitigating

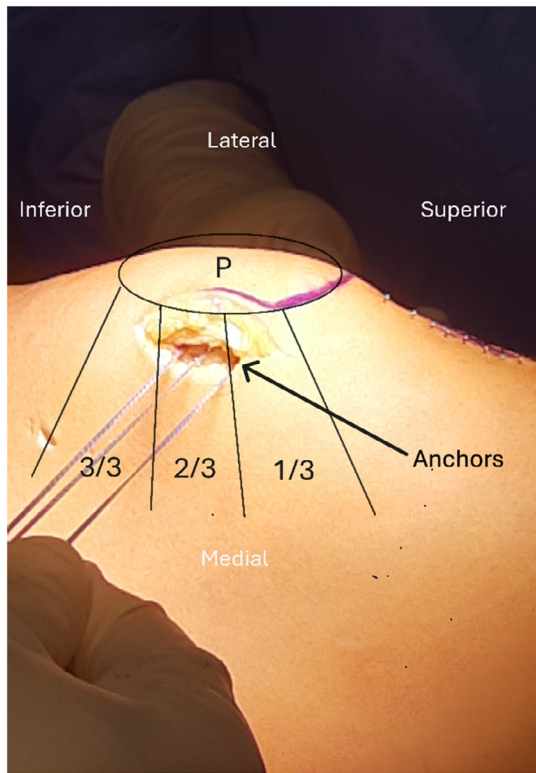


FIGURE 4 Intraoperative photograph of the right knee (supine position), showing the position of the suture anchors on the P. (P, patella.)

postoperative weakness and shortening rehabilitation compared with classical quadriceps tendon harvesting and possibly leading to less patellofemoral arthrosis.¹²

Potential complications of rectus femoris autograft harvesting include quadriceps weakness, inadvertent joint penetration, hematoma, and thigh pain.^{9,12} However, the incidence of these complications is expected to be lower than conventional quadriceps autograft harvesting.¹² Importantly, this technique also avoids compromising valgus stability of the knee or weakening knee flexors, risks more commonly encountered with hamstring autograft harvesting.¹¹ Advantages and disadvantages of the rectus femoris autograft for MPFL reconstruction are summarized in Table 2.

In summary, double-bundle MPFL reconstruction with a rectus femoris tendon autograft appears to be a promising technique that provides a robust graft with low donor-site morbidity. Additionally, it preserves the hamstrings for potential future ligament reconstruction, making it a particularly attractive option in select clinical scenarios.

DISCLOSURES

The authors (S.N., M.P.) declare that they have no known competing financial interests or personal

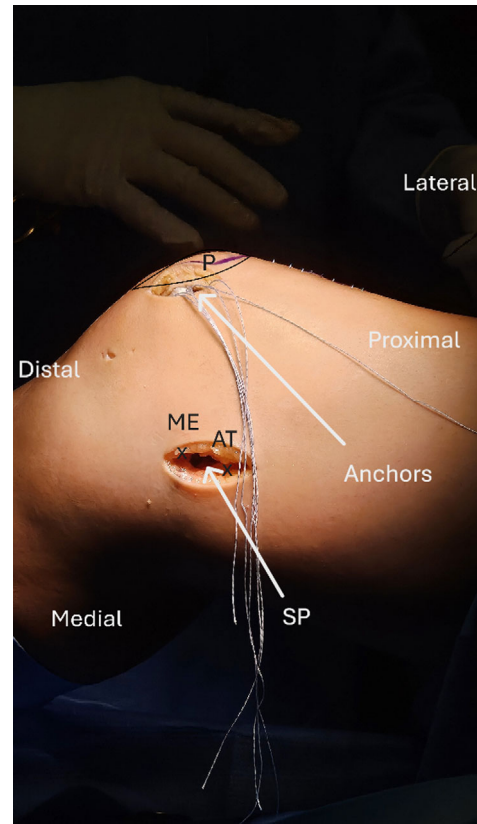


FIGURE 5 Intraoperative photograph of the right knee (supine position), showing SP. (AT, adductor tubercle; ME, medial epicondyle; P, patella; SP, Schöttle's point.)

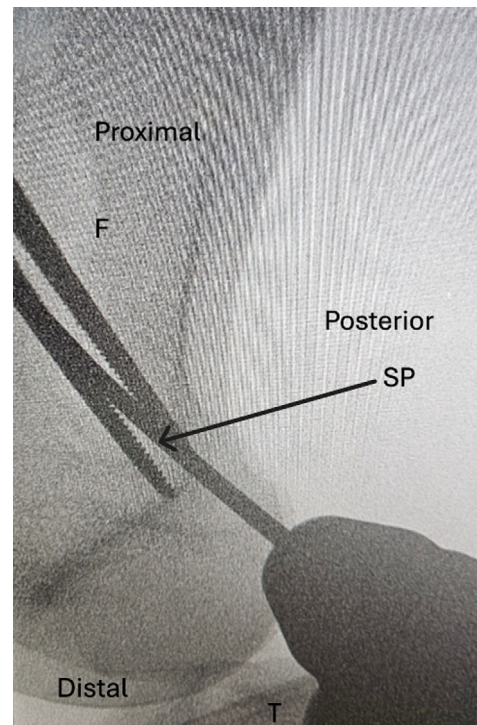


FIGURE 6 Intraoperative fluoroscopic image confirming correct identification of SP. (F, femur; SP, Schöttle's point; T, tibia.)

TABLE 1 Pearls and Pitfalls of MPFL Reconstruction Using Rectus Femoris Autograft

Pearls	Pitfalls
Make the skin incision higher than usual, 3-4 cm proximal to the superior pole of the patella	Incorrect incision placement may increase the risk of iatrogenic damage
Harvest only the superficial portion of the rectus femoris tendon to minimize donor-site morbidity	Excessive depth during harvest may inadvertently communicate with the knee joint
Maintain graft elasticity during harvest	Overtensioning or abrupt stripper advancement may cause graft rupture or inadequate length
Ensure a minimum graft length of 20 cm to allow tension-free fixation	Insufficient graft length compromises fixation
Identify Schöttle's point under fluoroscopy for anatomical placement	Malpositioned femoral tunnel can result in abnormal graft isometry
Tension graft with the knee between 45° and 60° flexion	Over- or undertensioning risks either overconstraint (leading to stiffness/pain) or persistent instability
MPFL, medial patellofemoral ligament.	

TABLE 2 Advantages and Disadvantages of Rectus Femoris Autograft for MPFL Reconstruction

Advantages	Disadvantages
Preserves hamstrings for possible future anterior cruciate ligament reconstruction	Potential for inadvertent joint penetration during harvest
Provides adequate graft length and strength with straightforward harvesting	Possible hematoma and thigh pain at harvest site
Harvesting only the superficial tendon reduces donor-site morbidity compared with classical quadriceps tendon grafts	Limited clinical experience; short-, mid-, and long-term outcomes not yet established
Lower risk of intra-articular contamination, reducing infection risk	Requires familiarity with graft-specific pearls to avoid technical complications
Maintains medial knee stability, unlike hamstring autograft harvest	Availability of graft may vary depending on patient anatomy
Potentially faster rehabilitation with less quadriceps weakening	
MPFL, medial patellofemoral ligament.	

relationships that could have appeared to influence the work reported in this article.

ORCID

Samo Novak  <https://orcid.org/0000-0002-4779-9054>
 Matic Povh  <https://orcid.org/0009-0001-5505-8176>

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